Step 1-3: Training Yolo + Gathering Metrics

Metrics with no training on a pretrained dataset

```
data = '../data/road_sign.yaml'
device = 'cuda'

# Metrics with no training with pretrained dataset
metrics = model.val(data=data)
```

Results

```
Ultralytics 8.4.23  Python-3.11.9 torch-2.5.1+cu121 CUDA:0 (NVIDIA
GeForce GTX 1660 SUPER, 6144MiB)
val: Fast image access  (ping: 0.00.0 ms, read: 1775.4326.6 MB/s,
size: 273.7 KB)
val: Scanning
C:\Users\matth\Documents\UNICODE\CompVis\lab6\data\labels\val.cache...
88 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━━━━━━━━━ 88/88
0.0s
                 Class     Images  Instances      Box(P          R
mAP50  mAP50-95): 100% ━━━━━━━━━━━━━━━━━━━━ 6/6 1.5it/s 4.0s0.2s1s
                   all         88        132    0.00673     0.0439
0.00366   0.000765
                person         10         19    0.00231      0.105
0.00144    0.00114
               bicycle         10         10          0          0
0          0
                   car         69         81    0.00136     0.0247
0.000702   0.000667
            motorcycle         16         22     0.0233     0.0455
0.0125    0.00125
Speed: 2.3ms preprocess, 3.8ms inference, 0.0ms loss, 2.3ms
postprocess per image
```

Metrics with 5 epochs of training on a pretrained dataset

```
# Use pretrained dataset and train on top of it (5 epochs)

epochs = 5

results = model.train(data=data, epochs=epochs, device=device);
model.val(data=data)
```

Results

```
Model summary (fused): 73 layers, 3,006,428 parameters, 0 gradients,
8.1 GFLOPs
                 Class     Images  Instances      Box(P          R
mAP50  mAP50-95): 100% ━━━━━━━━━━━━━━━━━━━━ 3/3 2.7it/s 1.1s0.7s
                   all         88        132      0.938      0.804
0.858      0.726
          trafficlight         10         19      0.895      0.451
0.611      0.474
                  stop         10         10          1      0.869
0.896       0.85
            speedlimit         69         81       0.93      0.986
0.986      0.846
             crosswalk         16         22      0.927      0.909
0.937      0.732
Speed: 0.2ms preprocess, 4.9ms inference, 0.0ms loss, 1.7ms
postprocess per image
```

Metrics with 5 epochs of training from scratch

```
# Yolo from scratch with 5 epochs

epochs = 5

model = YOLO("yolov8n.yaml")
model.train(data=data, epochs=epochs, device=device)
model.val(data=data)
```

Results

```
YOLOv8n summary (fused): 73 layers, 3,006,428 parameters, 0 gradients,
8.1 GFLOPs
                 Class     Images  Instances      Box(P          R
mAP50  mAP50-95): 100% ━━━━━━━━━━━━━━━━━━━━ 3/3 2.9it/s 1.0s0.7s
                   all         88        132       0.66      0.273
0.286      0.134
          trafficlight         10         19          1          0
0.00332   0.000822
                  stop         10         10      0.639      0.184
0.367      0.133
            speedlimit         69         81      0.313      0.605
0.496      0.262
             crosswalk         16         22       0.69      0.304
0.278      0.139
Speed: 0.2ms preprocess, 4.7ms inference, 0.0ms loss, 1.6ms
postprocess per image
```

Metrics with 15 epochs of training from scratch

```
# Yolo from scratch with 15 epochs

epochs = 15

model = YOLO("yolov8n.yaml")
model.train(data=data, epochs=epochs, device=device)
model.val(data=data)
```

Results

```
YOLOv8n summary (fused): 73 layers, 3,006,428 parameters, 0 gradients,
8.1 GFLOPs
                 Class     Images  Instances      Box(P          R
mAP50  mAP50-95): 100% ━━━━━━━━━━━━━━━━━━━━ 3/3 1.2it/s 2.6s1.7s
                   all         88        132      0.984      0.615
0.668      0.511
          trafficlight         10         19          1          0
0.056     0.0237
                  stop         10         10      0.986        0.9
0.896       0.76
            speedlimit         69         81      0.949      0.913
0.969      0.791
             crosswalk         16         22          1      0.649
0.752      0.472
Speed: 0.5ms preprocess, 8.2ms inference, 0.0ms loss, 3.8ms
postprocess per image
```

Step 4: Construct a table that compares the performance obtained under various settings.

Model	Pretrained?	Epochs	mAP50 (precision score)	mAP50-95 (strict precision score)	Thoughts
YOLOv8n	Yes	0	0.00366 (4th)	0.000765 (4th)	Failing detection. Using pretraining to try and find signs when the original model was not made for it will yield no sign detection.
YOLOv8n	Yes	5	0.858 (1st)	0.726(1st)	Very strong detection makes sense by using pretrained data along with our added training.
YOLOv8n	No	5	0.286 (3rd)	0.134 (3rd)	Weak detection, 5 epochs from scratch isn't enough to get consistent detection.
YOLOv8n	No	15	0.668 (2nd)	0.511 (2nd)	Moderate to Good detection, 15 epochs of sign training seems to be efficient to get good detection.

Step 5: Pick a small subset of test images and show qualitative results under various settings

Image 1:

Pre-Trained (No Training)	Pre-Trained (5 epochs of training data with stop signs)	From Scratch (5 epochs of training data with stop signs)	From Scratch (15 epochs of training data with stop signs)

Image 2:

Pre-Trained (No Training)	Pre-Trained (5 epochs of training data with stop signs)	From Scratch (5 epochs of training data with stop signs)	From Scratch (15 epochs of training data with stop signs)

Image 3:

Pre-Trained (No Training)	Pre-Trained (5 epochs of training data with stop signs)	From Scratch (5 epochs of training data with stop signs)	From Scratch (15 epochs of training data with stop signs)

Image 4:

Pre-Trained (No Training)	Pre-Trained (5 epochs of training data with stop signs)	From Scratch (5 epochs of training data with stop signs)	From Scratch (15 epochs of training data with stop signs)

Image 5:

Pre-Trained (No Training)	Pre-Trained (5 epochs of training data with stop signs)	From Scratch (5 epochs of training data with stop signs)	From Scratch (15 epochs of training data with stop signs)

			crosswalk